# SURAJ PRASAD KALAUNI

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## EDUCATION

Delhi Technological University, Delhi, India — *Aug 2024 – Aug 2026*
*M.Tech Software Engineering, GPA: 8.55* – Thesis: Machine Learning-Based Network Intrusion Detection Systems

Dr. A.P.J Abdul Kalam Technological University, Lucknow, India — *Jun 2020 – Mar 2024*
*B.Tech Computer Science and Engineering, GPA: 8.12*

## SKILLS

- **Languages**: Go (Golang), Java, C/C++, Python, SQL
- **Cloud & Containers**: Azure (AI Search), AWS (EC2, S3, Lambda, RDS, EMR), Docker, Kubernetes
- **Distributed Systems**: Microservices, Fault Tolerance, Scalability, Data Pipelines, Apache Spark.
- **Backend & Databases**: REST APIs, Spring Boot, FastAPI, PostgreSQL, MySQL, MongoDB, Redis.
- **Engineering Practices**: SDLC, Design Patterns, Object-Oriented Design, Data Structures & Algorithms, Unit & Integration Testing, CI/CD, Git, Debugging, Agile.
- **Security**: Data Protection, Encryption, Digital Signatures, JWT, Multi-Factor Authentication, RBAC, Intrusion & Anomaly Detection.

## WORK EXPERIENCE

**Samsung Research and Development | Software Engineer Intern** — Jul 2025 – Jan 2026

- Architected scalable AWS pipelines (EC2, S3, Lambda) for multi-terabyte network datasets; containerized services with Docker and Kubernetes, sustaining 99.9% uptime.
- Developed Python backend services for large-scale log analysis; improved root-cause identification by 20% and reduced latency by 15%.
- Automated 3+ data workflows and refactored legacy programs, eliminating recurring manual effort.
- Owned features across the SDLC, from requirements and sprint estimation through design, testing, and REST API release, using code reviews and CI.
- Designed and deployed a critical microservice, reducing system latency by 15% and securing a Pre-Placement Offer for a Senior Software Engineer.

## PROJECTS

**Fault-Tolerant Cloud Task Processing Service | Python, AWS, Docker**

- Engineered a cloud-native Python platform with REST APIs for 1,000+ asynchronous tasks, decoupling long-running jobs from client requests.
- Designed durable task state and automatic retries, enabling interrupted jobs to resume and achieving a 99% completion rate.
- Containerized the platform with Docker for AWS deployments; added structured logging and performance monitoring for faster failure diagnosis.

**Enterprise Support Agent on Azure | Java, Spring Boot, Azure AI Search, Docker**

- Built a retrieval-augmented LLM agent with LangChain and Azure AI Search to index enterprise documents and ground responses in retrieved context.
- Automated ticket creation and status updates; added multi-turn memory and prompt versioning for traceable releases.
- Integrated the agent with a Java Spring Boot backend and shipped containers through a CI/CD pipeline for consistent releases.

**Log Analysis & Anomaly Detection System | Spark Streaming, AWS EMR, TensorFlow**

- Built a real-time Spark Streaming ingestion layer on AWS EMR, processing 10M+ system log entries daily with under 3 s latency.
- Distributed stream computation across a scalable multi-node cluster, allowing capacity to grow with continuous log traffic.
- Trained a TensorFlow LSTM model on sequential log data to detect unusual network patterns, reducing false positives by 28%.

## PUBLICATIONS

- **Network Intrusion Detection for Vulnerable Networks Using CICIDS2017: Preprocessing, SMOTE Balancing, and Comparative Evaluation of Random Forest, XGBoost and Neural Network**, *Lecture Notes in Networks and Systems, ICICC 2026*, Springer Nature, 2026. DOI
- **Retinal Fundus Image and Deep Learning: A Novel Approach for Cardiovascular Risk Prediction**, *AIP Conference Proceedings*, 2025. Achieved **79.41% accuracy**. DOI

## ACHIEVEMENTS

- GATE CSE 2024: Qualified GATE CSE 2024 with Marks: 40.48, Score: 490/1000, Rank: 5943 among 123967.
- GATE DA 2024: Qualified GATE DA 2024 with Marks: 38, Score: 362/1000, Rank: 5973 among 39210.